Skill - 1

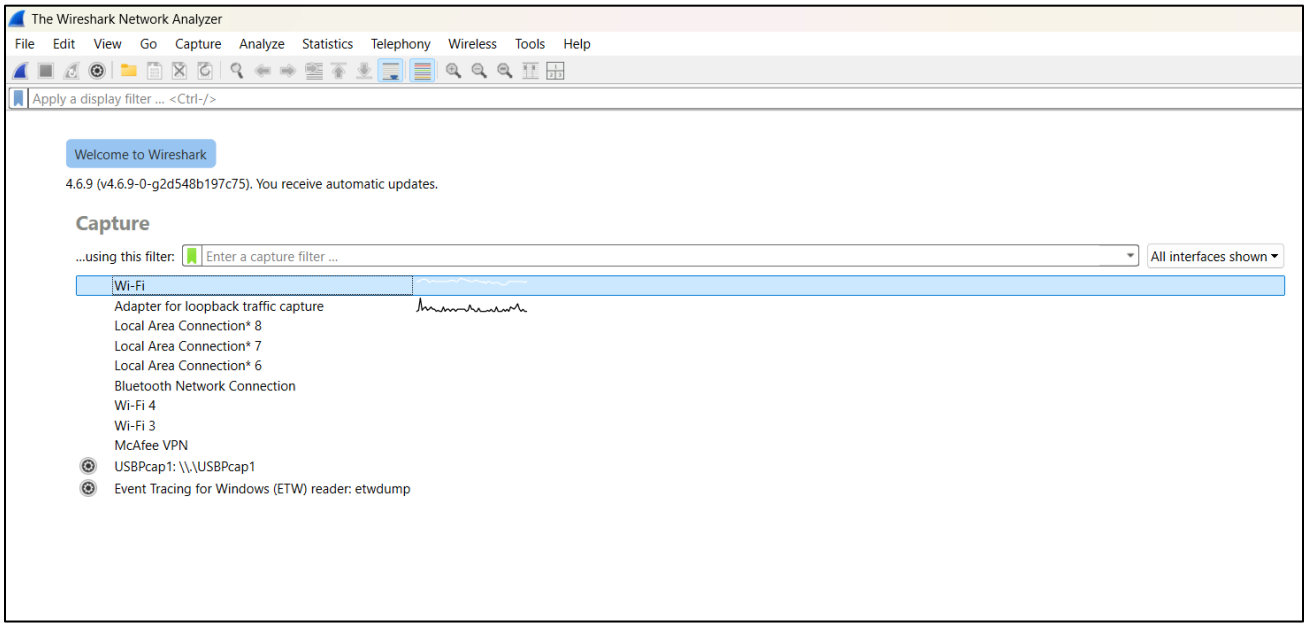
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Step 1: Install Wireshark

1. Open the browser.
2. Install Wireshark [Wireshark • Go Deep | Download]
3. Install it using the default settings.
4. Open Wireshark.

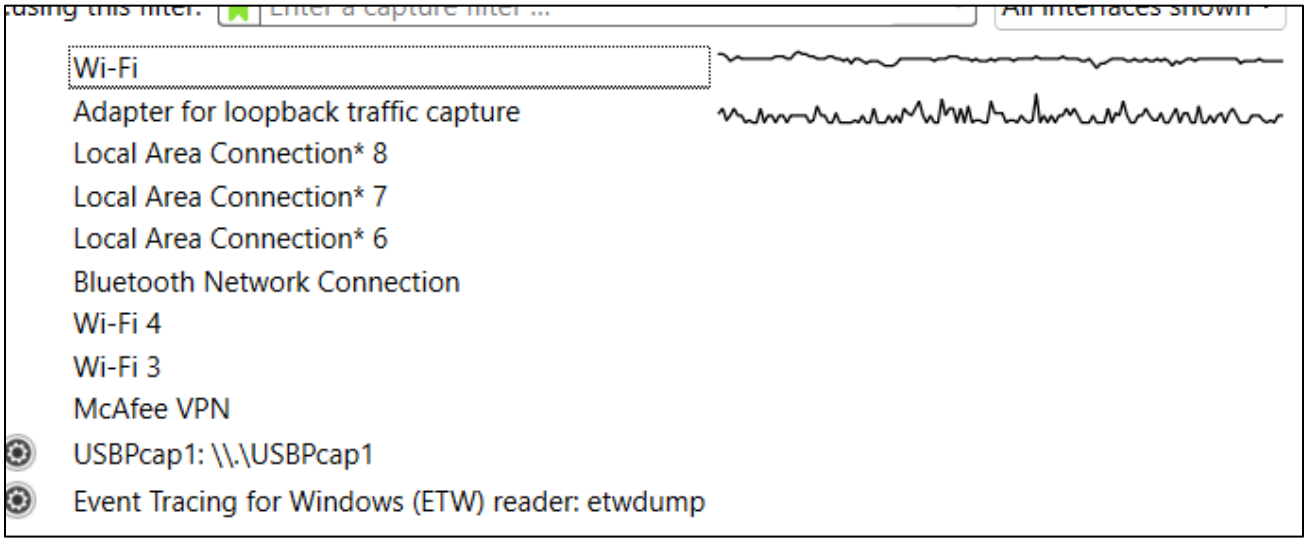


Step 2: Select your network interface

When Wireshark opens, you will see interfaces such as:

Wi-Fi, Ethernet, Bluetooth

1. If you are connected through Wi-Fi, select Wi-Fi.
2. Double-click the Wi-Fi interface.
3. Packet capture will start.



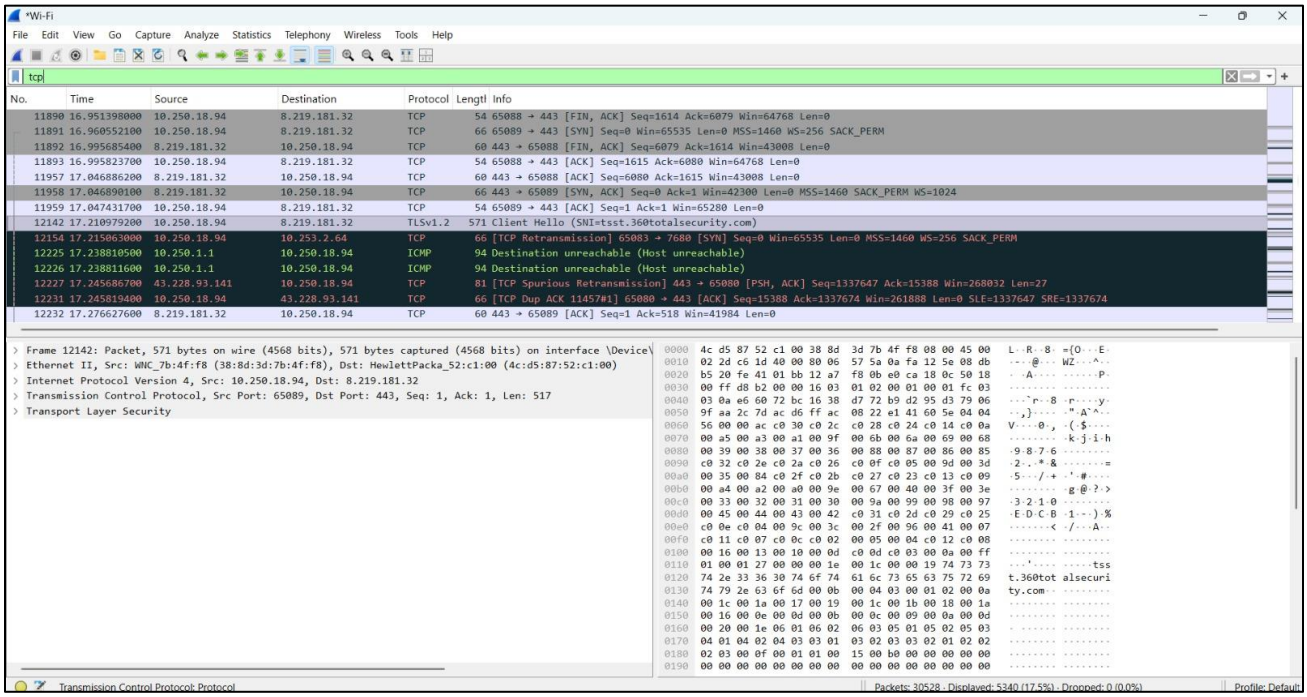
Step 3: Generate network traffic

While Wireshark is capturing:

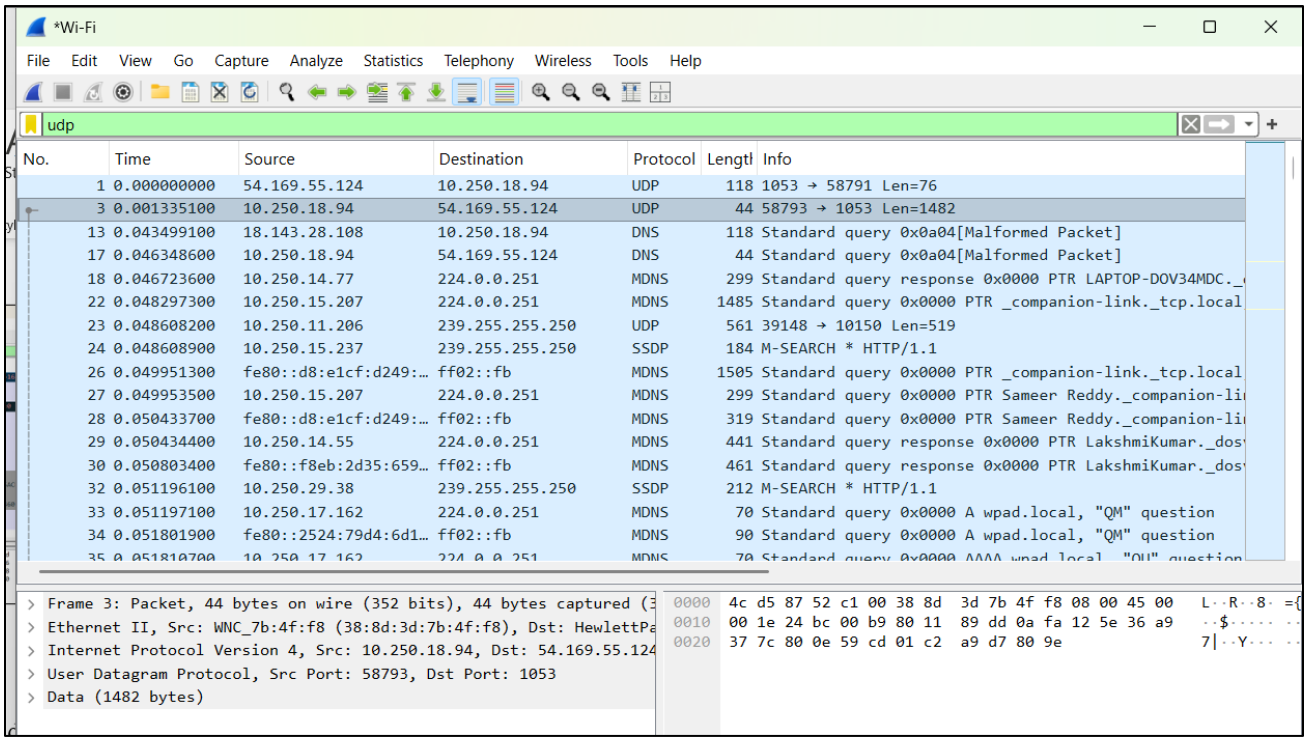
1. Open a new browser tab.
2. Visit a website such as https://www.google.com.
3. Open two or three different websites.
4. Wait for approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button.

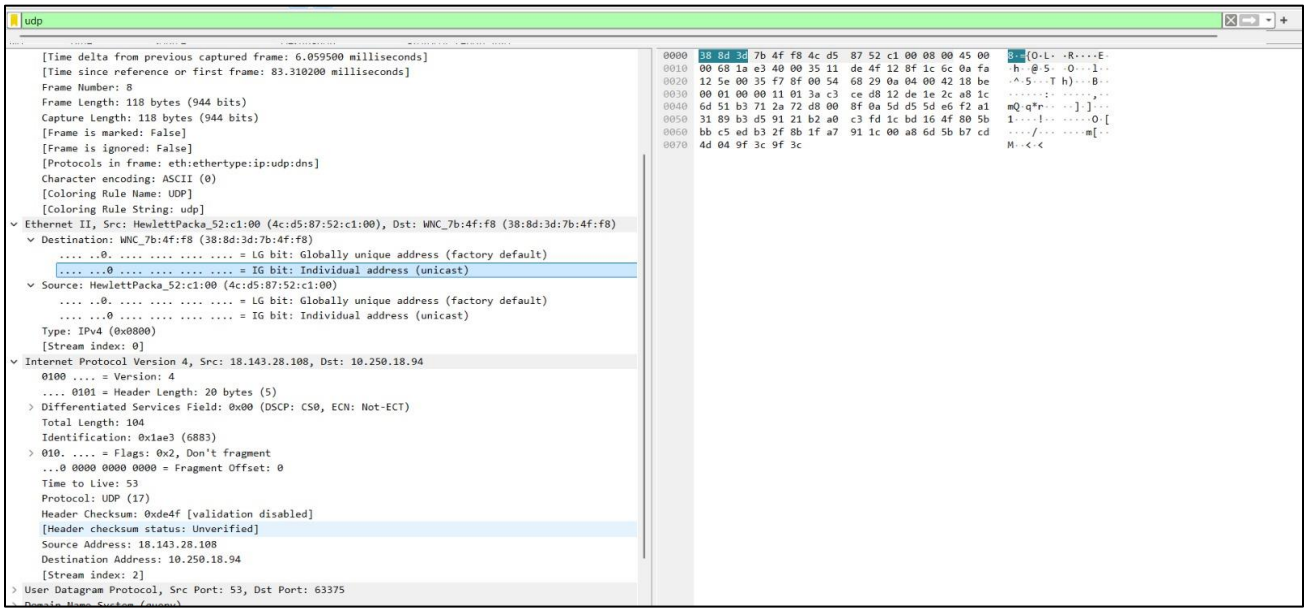
You should now see captured packets.

1.TCP

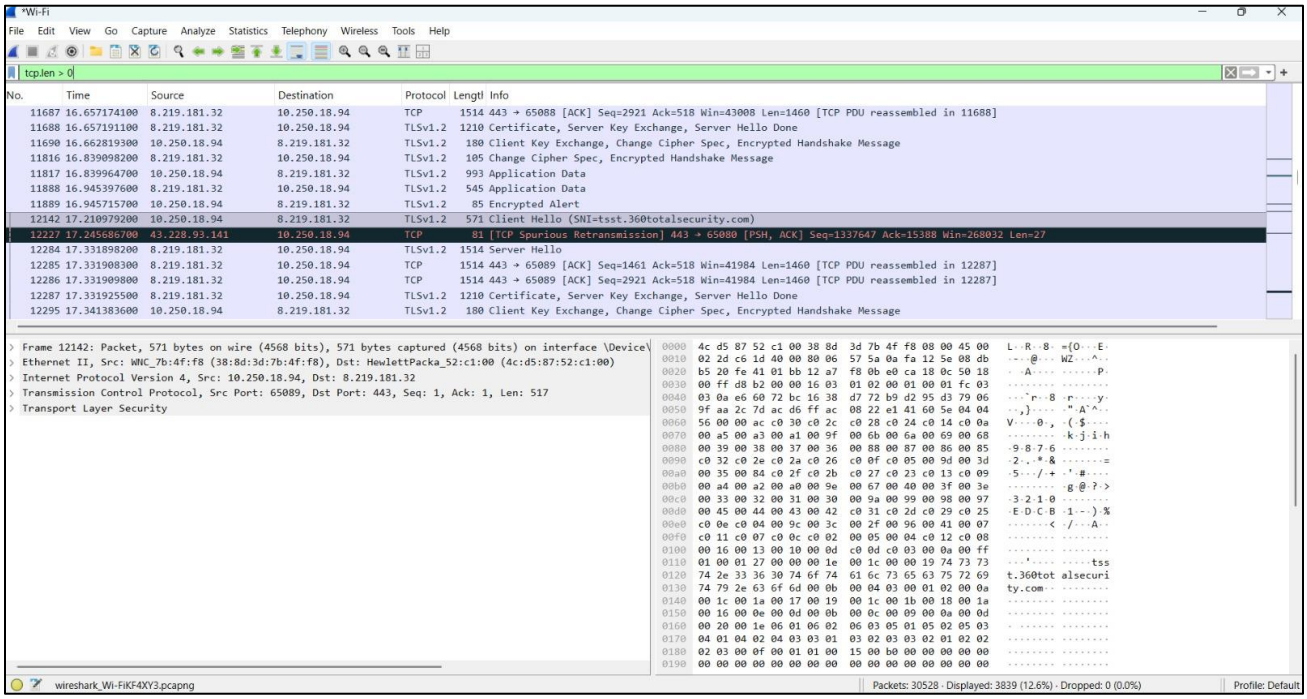

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
11890 16.951398000 10.250.18.94 8.219.181.32 TCP 54 65088 → 443 [FIN, ACK] Seq=1614 Ack=6079 Win=64768 Len=0
11891 16.960552100 10.250.18.94 8.219.181.32 TCP 66 65089 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11892 16.995685400 8.219.181.32 10.250.18.94 TCP 60 443 → 65088 [FIN, ACK] Seq=6079 Ack=1614 Win=43008 Len=0
11893 16.995823700 10.250.18.94 8.219.181.32 TCP 54 65088 → 443 [ACK] Seq=1615 Ack=6080 Win=64768 Len=0
11957 17.046886200 8.219.181.32 10.250.18.94 TCP 60 443 → 65088 [ACK] Seq=6080 Ack=1615 Win=43008 Len=0
11958 17.046890100 8.219.181.32 10.250.18.94 TCP 66 443 → 65089 [SYN, ACK] Seq=0 Ack=1 Win=42300 Len=0 MSS=1460 SACK_PERM WS=1024
11959 17.047431700 10.250.18.94 8.219.181.32 TCP 54 65089 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
12142 17.210979200 10.250.18.94 8.219.181.32 TLSv1.2 571 Client Hello (SNI=tsst.360totalsecurity.com)
12154 17.215063000 10.250.18.94 10.253.2.64 TCP 66 [TCP Retransmission] 65083 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
12225 17.238810500 10.250.1.1 10.250.18.94 ICMP 94 Destination unreachable (Host unreachable)
12226 17.238811600 10.250.1.1 10.250.18.94 ICMP 94 Destination unreachable (Host unreachable)
12227 17.245686700 43.228.93.141 10.250.18.94 TCP 81 [TCP Spurious Retransmission] 443 → 65080 [PSH, ACK] Seq=1337647 Ack=15388 Win=268032 Len=27
12231 17.245819400 10.250.18.94 43.228.93.141 TCP 66 [TCP Dup ACK 11457#1] 65080 → 443 [ACK] Seq=15388 Ack=1337674 Win=261888 Len=0 SLE=1337647 SRE=1337674
12232 17.276627600 8.219.181.32 10.250.18.94 TCP 60 443 → 65089 [ACK] Seq=1 Ack=518 Win=41984 Len=0
Frame 12142: Packet, 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interface \Device\
Ethernet II, Src: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Internet Protocol Version 4, Src: 10.250.18.94, Dst: 8.219.181.32
Transmission Control Protocol, Src Port: 65089, Dst Port: 443, Seq: 1, Ack: 1, Len: 517
Transport Layer Security
Transmission Control Protocol: Protocol
Packets: 30528 · Displayed: 5340 (17.5%) · Dropped: 0 (0.0%)
Profile: Default

2.UDP

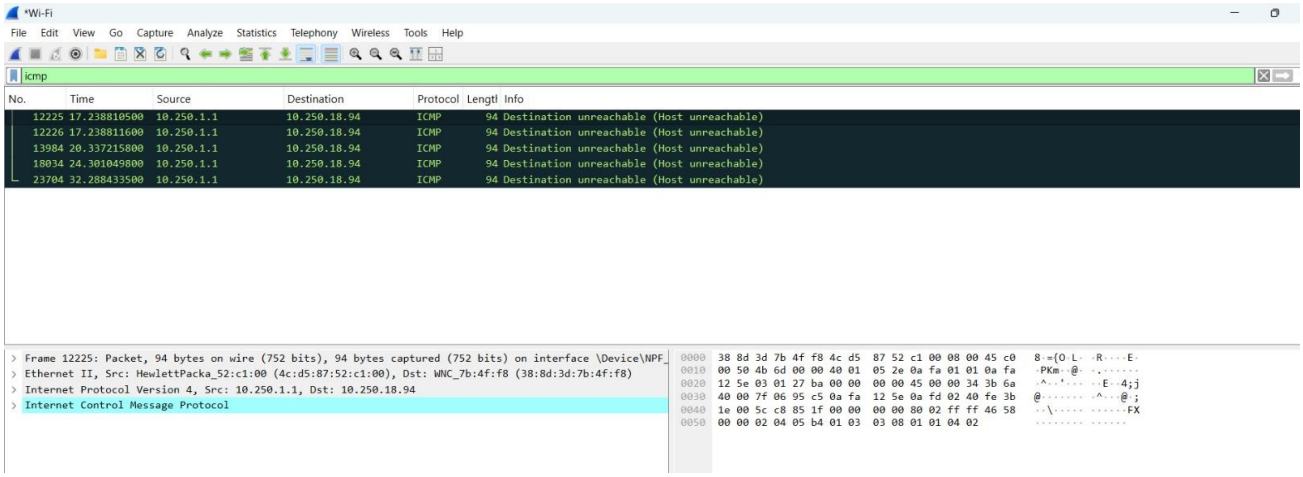

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Lengtl Info
1 0.000000000 54.169.55.124 10.250.18.94 UDP 118 1053 → 58791 Len=76
3 0.001335100 10.250.18.94 54.169.55.124 UDP 44 58793 → 1053 Len=1482
13 0.043499100 18.143.28.108 10.250.18.94 DNS 118 Standard query 0x0a04[Malformed Packet]
17 0.046348600 10.250.18.94 54.169.55.124 DNS 44 Standard query 0x0a04[Malformed Packet]
18 0.046723600 10.250.14.77 224.0.0.251 MDNS 299 Standard query response 0x0000 PTR LAPTOP-DOV34MDC._
22 0.048297300 10.250.15.207 224.0.0.251 MDNS 1485 Standard query 0x0000 PTR _companion-link._tcp.local
23 0.048608200 10.250.11.206 239.255.255.250 UDP 561 39148 → 10150 Len=519
24 0.048608900 10.250.15.237 239.255.255.250 SSDP 184 M-SEARCH * HTTP/1.1
26 0.049951300 fe80::d8:e1cf:d249:… ff02::fb MDNS 1505 Standard query 0x0000 PTR _companion-link._tcp.local
27 0.049953500 10.250.15.207 224.0.0.251 MDNS 299 Standard query 0x0000 PTR Sameer Reddy._companion-li
28 0.050433700 fe80::d8:e1cf:d249:… ff02::fb MDNS 319 Standard query 0x0000 PTR Sameer Reddy._companion-li
29 0.050434400 10.250.14.55 224.0.0.251 MDNS 441 Standard query response 0x0000 PTR LakshmiKumar._dos
30 0.050803400 fe80::f8eb:2d35:659… ff02::fb MDNS 461 Standard query response 0x0000 PTR LakshmiKumar._dos
32 0.051196100 10.250.29.38 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
33 0.051197100 10.250.17.162 224.0.0.251 MDNS 70 Standard query 0x0000 A wpad.local, "QM" question
34 0.051801900 fe80::2524:79d4:6d1… ff02::fb MDNS 90 Standard query 0x0000 A wpad.local, "QM" question
Frame 3: Packet, 44 bytes on wire (352 bits), 44 bytes captured (3
Ethernet II, Src: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8), Dst: HewlettPa
Internet Protocol Version 4, Src: 10.250.18.94, Dst: 54.169.55.124
User Datagram Protocol, Src Port: 58793, Dst Port: 1053
Data (1482 bytes)
0000 4c d5 87 52 c1 00 38 8d 3d 7b 4f f8 08 00 45 00
0010 00 1e 24 bc 00 b9 80 11 89 dd 0a fa 12 5e 36 a9
0020 37 7c 80 0e 59 cd 01 c2 a9 d7 80 9e


udp
[Time delta from previous captured frame: 6.059500 milliseconds]
[Time since reference or first frame: 83.310200 milliseconds]
Frame Number: 8
Frame Length: 118 bytes (944 bits)
Capture Length: 118 bytes (944 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8)
Destination: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 18.143.28.108, Dst: 10.250.18.94
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 104
Identification: 0x1ae3 (6883)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 53
Protocol: UDP (17)
Header Checksum: 0xde4f [validation disabled]
[Header checksum status: Unverified]
Source Address: 18.143.28.108
Destination Address: 10.250.18.94
[Stream index: 2]
User Datagram Protocol, Src Port: 53, Dst Port: 63375

3.TCP.LEN > 0


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len > 0
No. Time Source Destination Protocol Lengtl Info
11687 16.657174100 8.219.181.32 10.250.18.94 TCP 1514 443 → 65088 [ACK] Seq=2921 Ack=518 Win=43008 Len=1460 [TCP PDU reassembled in 11688]
11688 16.657191100 8.219.181.32 10.250.18.94 TLSv1.2 1210 Certificate, Server Key Exchange, Server Hello Done
11690 16.662819300 10.250.18.94 8.219.181.32 TLSv1.2 180 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
11816 16.839098200 8.219.181.32 10.250.18.94 TLSv1.2 105 Change Cipher Spec, Encrypted Handshake Message
11817 16.839964700 10.250.18.94 8.219.181.32 TLSv1.2 993 Application Data
11888 16.945397600 8.219.181.32 10.250.18.94 TLSv1.2 545 Application Data
11889 16.945715700 10.250.18.94 8.219.181.32 TLSv1.2 85 Encrypted Alert
12142 17.210979200 10.250.18.94 8.219.181.32 TLSv1.2 571 Client Hello (SNI=tsst.360totalsecurity.com)
12227 17.245686700 43.228.93.141 10.250.18.94 TCP 81 [TCP Spurious Retransmission] 443 → 65080 [PSH, ACK] Seq=1337647 Ack=15388 Win=268032 Len=27
12284 17.331898200 8.219.181.32 10.250.18.94 TLSv1.2 1514 Server Hello
12285 17.331908300 8.219.181.32 10.250.18.94 TCP 1514 443 → 65089 [ACK] Seq=1461 Ack=518 Win=41984 Len=1460 [TCP PDU reassembled in 12287]
12286 17.331909800 8.219.181.32 10.250.18.94 TCP 1514 443 → 65089 [ACK] Seq=2921 Ack=518 Win=41984 Len=1460 [TCP PDU reassembled in 12287]
12287 17.331925500 8.219.181.32 10.250.18.94 TLSv1.2 1210 Certificate, Server Key Exchange, Server Hello Done
12295 17.341383600 10.250.18.94 8.219.181.32 TLSv1.2 180 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
Frame 12142: Packet, 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interface \Device\
Ethernet II, Src: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Internet Protocol Version 4, Src: 10.250.18.94, Dst: 8.219.181.32
Transmission Control Protocol, Src Port: 65089, Dst Port: 443, Seq: 1, Ack: 1, Len: 517
Transport Layer Security
wireshark_Wi-FiKF4XY3.pcapng
Packets: 30528 · Displayed: 3839 (12.6%) · Dropped: 0 (0.0%)
Profile: Default

4. ICMP



Step 4: Apply display filters

In the Wireshark filter bar, enter each filter separately and press Enter.

Filter	**Purpose**
tcp	Displays TCP packets
udp	Displays UDP packets
icmp	Displays ICMP packets
tcp.len > 0	Displays TCP packets carrying data

First, enter: TCP

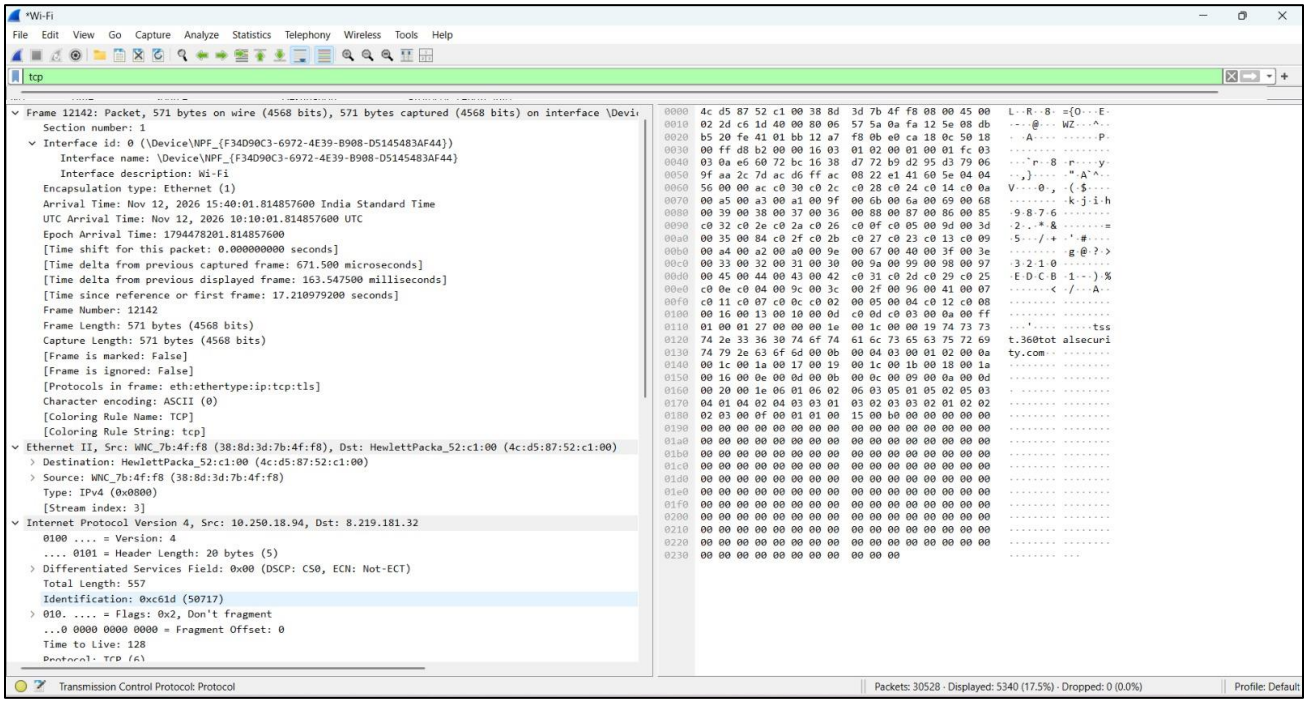
Observe the packets displayed.

Then enter: UDP

Observe the UDP packets.

Step 5: Inspect packet headers
Select any TCP packet from the packet list.

In the packet details pane, expand these sections:
1. Frame
2. Ethernet II
3. Internet Protocol Version 4
4. Transmission Control Protocol



Record the following values:

Parameter	Where to find it
Frame length	Frame
Ethernet Header	Ethernet II
Source MAC Address	Ethernet II
Destination MAC Address	Ethernet II
IPv4 Header Length	Internet Protocol Version 4
IPv4 Total Length	Internet Protocol Version 4
TCP Header Length	Transmission Control Protocol
TCP Segment Length	Transmission Control Protocol

Repeat the process for one UDP packet. Observe the UDP header and its length field.

Step 6: Calculate header overhead

For a basic Ethernet + IPv4 + TCP packet without TCP options:

1. **TCP Packet Calculation**

From the Wireshark capture:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- TCP Header = 20 bytes
- TCP Payload = 517 bytes

Total Packet Size:
14 + 20 + 20 + 517 = 571 bytes
Total Overhead:
14 + 20 + 20 = 54 bytes
Payload Efficiency:
(517 / 571) × 100 = **90.54%**

2. UDP Packet Calculation

From the Wireshark capture:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- UDP Header = 8 bytes
- UDP Payload = 76 bytes

Total Packet Size:
14 + 20 + 8 + 76 = 118 bytes
Total Overhead:
14 + 20 + 8 = 42 bytes
Payload Efficiency:
(76 / 118) × 100 = **64.41%**

Note: Wireshark may not capture or display the Ethernet FCS. Use the actual captured packet fields when recording your observations.

Step 7:
Fill in your observation table

Record values from your own Wireshark capture.

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	571 bytes	118 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	20 bytes
IP Total Length	557 bytes	104 bytes
Total Payload Length	517 bytes	76 bytes

Total Overhead	54 bytes	42 bytes
Payload Efficiency	90.54%	64.41%

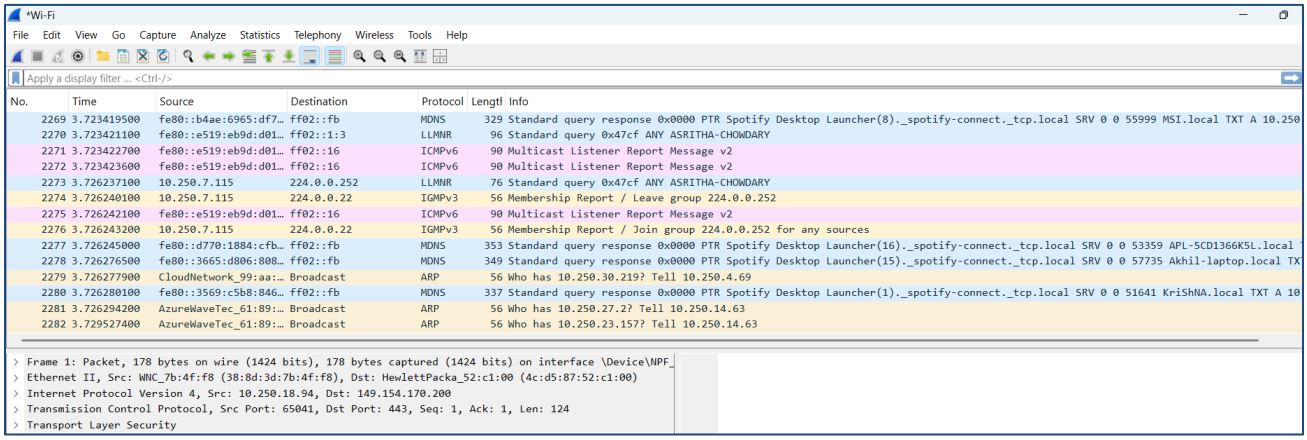
Steps to save a capture file

Step 1: Start capturing packets

1. Open Wireshark.
2. Select your Wi-Fi or Ethernet interface.
3. Double-click the interface to start capturing packets.

Step 2: Stop the capture

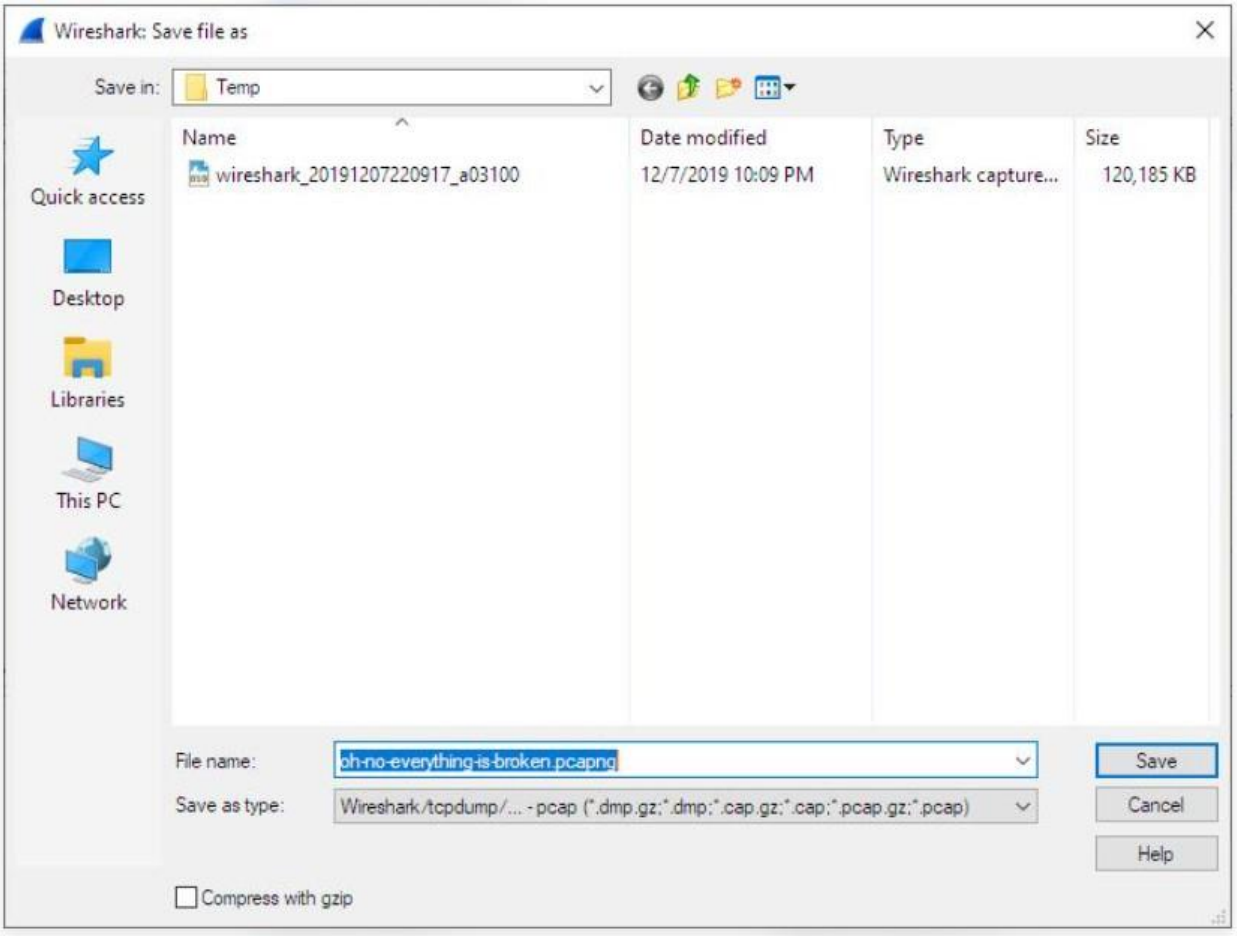
Once you have generated sufficient network traffic:
1. Click the red Stop button (top-left corner).
2. Your captured packets will remain visible on the screen.



Step 3: Save the capture

1. Click File in the menu.
2. Select Save As.
3. Choose the folder where you want to save the file.
4. Enter a filename.
5. Keep the file format as pcapng.
6. Click Save.

For example: sender_capture.pcapng



How to open a saved capture file

1. Open Wireshark.

2. Click File → Open.

3. Select your .pcapng or .pcap file.

4. Click Open.

5. Enter icmp in the display filter bar to view ICMP packets.